SpeechLP

SpeechLP®

Research Foundation & Logic Model

ESSA Tier 4 “Demonstrates a Rationale” Evidence Packet



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NOVEMBER 2026

RESEARCH-BASED

DESIGN

ESSA IV

Logic model & research base

DEMONSTRATES A RATIONALE

ESSA Tier 4 Evidence Statement for SpeechLP

SpeechLP

This packet documents how SpeechLP's design is grounded in published research on speech sound development and intervention, describes the program's features and intended outcomes in a logic model, and outlines a planned study of its effects. Together these components address the Every Student Succeeds Act (ESSA) criteria for Tier 4, "**Demonstrates a Rationale**."

TO COMPLETE — Independent review. The OgStar packet includes a certificate signed by an independent research organization (LXD Research). SpeechLP does not have one yet, so this page is a self-attestation. Once a third-party reviewer (e.g., LXD Research, Digital Promise, ISTE, or a university partner) has evaluated this packet, replace this box with their recognition statement, reviewer name/title, signature, and review date. Do not add a reviewer seal or signature until the review is actually issued.

Reviewer name, credentials Organization	**DATE**

Executive Summary

SpeechLP is a mobile app, built by practicing speech-language pathologists, that gives children frequent, structured, and measured practice of speech sounds at home and at school. Through its companion SLP tool, SpeechTherapist.app, therapists assign that practice and see how it is going.

What SpeechLP does

An age-based screener of about 15 words per level identifies a child's tricky sounds. Short video lessons, in which two certified SLPs model each sound, are followed by speaking games that draw on more than 1,000 target words across all 24 English consonants. Automated speech recognition gives feedback on every word, and progress is reported to parents, teachers, and SLPs.

Who it serves

Preschool and elementary-age children who are working on speech sounds, whether or not they have a diagnosed speech sound disorder. Families and SLPs use it alongside therapy; it does not replace assessment or treatment by a licensed SLP. In the initial study, most children were recorded as ages 2–8.

Why it should work

Speech sound disorders are common, affecting an estimated 2.1%–23% of children aged 4–6, and are linked to later literacy difficulties. Treatment is effective, and research consistently points to practice amount, frequency, and feedback as drivers of progress. Yet therapy time is limited and home practice is rarely measured. SpeechLP is designed to increase practice and make it visible (see the Research Summary and Logic Model).

Initial evidence

A descriptive retrospective study of 101 returning children found that app-scored accuracy on the same sounds rose from 69.4% to 74.9% between two consecutive months (+5.5 percentage points, 95% CI 0.4 to 10.4). The scoring is not yet validated against SLP judgment, and there was no comparison group, so this is not evidence of effectiveness. It is a starting point for the studies that follow (see Initial Evidence).

ESSA status

This packet documents a research-based rationale and logic model and a completed initial study, with prospective evaluation planned, meeting the criteria for **ESSA Tier 4, "Demonstrates a Rationale."** Independent third-party review is pending.

Evidence goals

1. **Validate scoring** against several SLP raters, acoustic measures, and parent judgments (Appendix A).
2. **Study home practice dose** using de-identified existing data.
3. **Test effectiveness** in prospective studies with a comparison group, building toward Tier 3 or Tier 2 evidence.

~15	1,000+	2	+5.5 pp
screener words per age level	target words, all 24 consonants	CCC-SLPs modeling every sound	app-scored accuracy, month 1 to 2 (n = 101)

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Understanding ESSA Evidence

Educators search for high-quality, evidence-based interventions to strengthen grant applications, to support comprehensive and targeted schools, and to implement new programming. Evidence requirements under the Every Student Succeeds Act (ESSA) are designed to help states, districts, and schools identify programs, practices, products, and policies that work across various populations.

Educational programs document their evidence of design, effectiveness, and impact in order to be eligible for federal funding. ESSA defines four levels of evidence: strong (Tier 1), moderate (Tier 2), and promising (Tier 3) evidence of effectiveness, or a program that demonstrates a rationale to be effective (Tier 4). The U.S. Department of Education's non-regulatory guidance describes Tier 4 as an activity with a well-specified logic model informed by research or evaluation, accompanied by ongoing efforts to examine its effects (U.S. Department of Education, 2016).

This packet addresses the requirements for Tier 4:

- ✓ Documentation of how the product's design relates to intended outcomes, with corresponding published academic research
- ✓ A logic model describing the product's resources, activities, outputs, and outcomes
- ✓ An initial descriptive study has been completed (Silver et al., 2026), and a prospective evaluation is planned (see "Planned Evaluation")
- ★ A third-party research organization reviews the documentation for ESSA validation [pending]

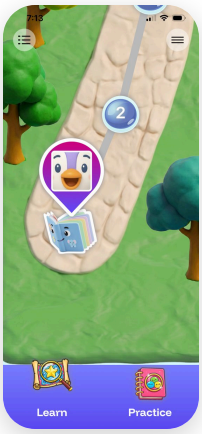


When speech-language tools are built on what is known about how children acquire and practice speech sounds, families, teachers, and SLPs can extend high-quality practice well beyond the therapy room. This packet maps each SpeechLP feature to that research and lays out how its effects will be measured.

TO COMPLETE: Replace with a quote from the independent reviewer once the review is issued.

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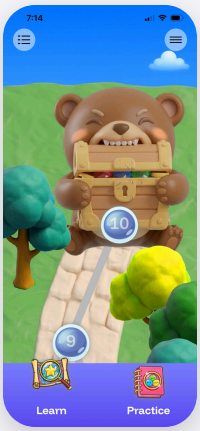
SpeechLP Supports Speech Sound Development

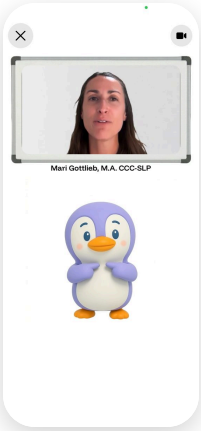


SpeechLP is a mobile app that helps children practice clear speech sounds between, or in addition to, sessions with a speech-language pathologist. A brief age-based screener of about 15 words per age level identifies which sounds are tricky for a child. The app then builds a guided learning path of short SLP-led video lessons and playful practice games drawn from a library of more than 1,000 target words. SLPs use the companion **SpeechTherapist.app** to assign practice to students and see their progress.

Screen, Then Practice What Matters

Children begin with a short screener matched to their age level, with about 15 words per level that sample the speech sounds expected for that age. Results determine which sounds the child practices, so time is spent on the sounds that need attention rather than on sounds already mastered.





Taught by Licensed SLPs, Practiced Through Play

Each step of the learning path opens with a short video lesson from a certified speech-language pathologist. The videos, created by Mari Gottlieb, M.A., CCC-SLP, feature both Mari and Alissa Fox, M.S., CCC-SLP, modeling each target sound, using cueing written by Alissa Fox. Children then practice their target sounds in words through games such as *Beary Hungry*, *Baby Talk*, *Chicken Little*, and *Pilot Penguin*.

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SpeechLP

User Voices

Parents and speech-language pathologists are the people who support children's practice day to day, and the best judges of whether a practice tool fits their lives and caseloads. This section shares their experiences with SpeechLP.

TO COMPLETE — Real testimonials only. Replace each placeholder below with a genuine quote, collected with written permission to publish. Use first name and role (e.g., "Jessica, SLP, Ohio").

C

[Carrie's quote: paste her exact words here.]
Carrie
SpeechLP parent, mom of a young child

[Quote from an SLP using SpeechLP to support their caseload, e.g., assigning home practice and reviewing progress in SpeechTherapist.app.]
Name
Speech-Language Pathologist, State

S

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SpeechLP® Foundational Research Summary

Introduction

Speech sound disorders are among the most common communication difficulties of childhood. In a large population-based sample, Shriberg, Tomblin, and McSweeny (1999) found that about 3.8% of 6-year-olds had a speech delay. ASHA reports estimated speech sound disorder prevalence of 2.1%–23% at ages 4–6 and persistent speech sound disorder in about 3.6% of children at age 8 (ASHA, n.d.). Children with speech sound disorders are at increased risk for later difficulties with reading and spelling, particularly when difficulties persist or co-occur with language impairment (Anthony et al., 2011), and residual speech errors in school-aged children have been linked to social, emotional, and academic impacts (Hitchcock et al., 2015).

Effective speech intervention depends heavily on practice. Children need many accurate productions of a target sound, distributed over time, to make lasting gains (Maas et al., 2008; Williams, 2012). Yet the amount of practice a child receives in typical school-based service—often short, small-group sessions once or twice a week—is limited (Sugden et al., 2018). Treatment works when it is delivered: a meta-analysis of controlled studies with children aged 3–18 found medium-to-high effects on phonological accuracy (Kunnari et al., 2024).

SpeechLP is a mobile app that extends speech sound practice beyond the therapy session. It combines a brief screener, SLP-led video lessons, a large library of target words, and game-based word practice, with progress reports for parents, teachers, and SLPs. SpeechLP is designed to supplement, not replace, assessment and intervention by a licensed speech-language pathologist.

Research on SpeechLP's Key Differentiating Features

1. Age-Based Screening of Developmentally Expected Sounds

Children acquire English consonants in a broadly predictable developmental sequence (Smit et al., 1990). A review of 15 U.S. studies covering 18,907 children found that most English consonants were acquired by age 5;0 under a 90% acquisition criterion, with /ɹ/, /θ/, /ð/, and /ʒ/ among the later-acquired sounds (Crowe & McLeod, 2020). These population norms are not individual diagnostic deadlines, but they indicate which sounds a child of a given age would typically be expected to produce. Identifying which sounds are in error is the starting point for selecting intervention targets (McLeod & Baker, 2017).

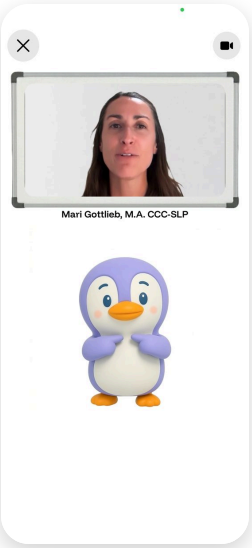
SpeechLP's screener is organized by age level, with about 15 words per level that sample the speech sounds children are expected to produce by that age (Crowe & McLeod, 2020). Results personalize the learning path so each child practices the specific sounds that are difficult for them. Screener results are shared in a report that parents can bring to their child's SLP or teacher, supporting earlier conversations about whether a full evaluation is warranted. The screener is not a diagnostic assessment and does not replace an evaluation by a licensed SLP.

TO COMPLETE: Add a screenshot of the screener and screener report (none were available for this draft), and briefly describe the age levels (e.g., ages 3, 4, 5, 6, 7+), how the words for each level were selected, and which developmental norms they follow. If the screener still samples initial, medial, and final positions, say so here.

2. Expert Modeling Through SLP-Led Video Lessons

Learning a new speech movement benefits from clear instruction and a model to imitate. Motor learning research applied to speech emphasizes pre-practice—explaining and demonstrating how a sound is made—before a child begins repeated practice (Maas et al., 2008). Video modeling is also a well-established, effective instructional technique for children (Bellini & Akullian, 2007).

Each step of the SpeechLP learning path opens with a brief video in which a certified speech-language pathologist explains and demonstrates the target sound. The videos were created by Mari Gottlieb, M.A., CCC-SLP, and feature both Mari and Alissa Fox, M.S., CCC-SLP, who provide the models; the cueing used to guide children's productions was written by Alissa Fox. Because lessons are recorded, every child receives a consistent, expert model, and families without easy access to an SLP still see correct placement and production.

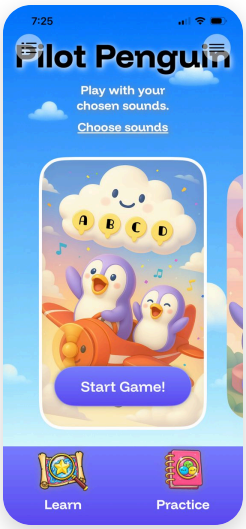


Each lesson begins with a certified SLP modeling the target sound on video, followed by a character guide who leads the child into practice.

3. High-Dose Practice From a Library of 1,000+ Words

The number of practice trials matters. Children with speech sound disorders who produce more targets per session tend to make greater gains (Edeal & Gildersleeve-Neumann, 2011), and systematic reviews identify treatment intensity as a key ingredient of effective speech intervention (Kaipa & Peterson, 2016). How practice is distributed also matters: in a randomized study of preschool children receiving the multiple oppositions approach, treatment three times weekly produced greater phonological gains than treatment once weekly when cumulative intervention intensity was held constant (Allen, 2013), and in children with childhood apraxia of speech, twice-weekly motor speech treatment for 10 weeks outperformed once-weekly treatment on articulation and functional communication (Namasivayam et al., 2015). Motor learning research adds that practice amount, distribution, variability, and feedback schedule may affect retention and transfer (Maas et al., 2008).

SpeechLP draws from a library of more than 1,000 words, organized by sound and word position, so children can produce many different exemplars of their target sound rather than repeating the same small set. Practice currently focuses on the word level. Treatment traditionally moves from words toward phrases, sentences, and connected speech (McLeod & Baker, 2017), and Speech Motor Chaining applies motor learning principles through structured practice that progresses from syllables to self-generated sentences (Preston et al., 2019). Phrase- and sentence-level practice is a planned next step for SpeechLP, informed by this research.



Children choose their target sounds, then practice them across many different words (Pilot Penguin).

4. Motivating, Purposeful Game-Based Practice

Repetitive drill can be hard for young children to sustain. A review of empirical studies found that gamification generally produces positive effects on engagement and motivation, with outcomes depending on the design and context (Hamari et al., 2014). Studies of tablet-based speech practice games for children have reported high engagement and production gains comparable to traditional paper-based practice (Hair et al., 2021), and computer-based approaches are a growing part of speech sound intervention (Furlong et al., 2017).

In SpeechLP, every game is built around speaking. In *Beary Hungry* the child feeds the bear by saying the word; in *Baby Talk* and *Chicken Little* the child must "say it right to make it pass." Children can choose which sounds to play with, progress along a visual map with numbered levels and rewards, and celebrate success with animated characters. These elements are designed to increase the number of productions a child is willing to attempt in a session.



Every game requires the child to produce their target sounds to advance.

Automated feedback on every word. Each practice card shows a pictured target word and plays a spoken model the child can listen to. Automated speech recognition then evaluates the child's production against the card's target sound, and a successful production advances the game (Silver et al., 2026). Automated feedback in structured speech practice is an active research area. In a single-case experiment, listener-rated /ɹ/ accuracy in practiced utterances was higher immediately after 30 minutes of AI-assisted Speech Motor Chaining practice than before it (Benway & Preston, 2024), and automated tools for analyzing children's speech are a growing field of study (McKechnie et al., 2018). In Swedish children aged 4–10, a speech recognition model rated single words within about 5 percentage points of an expert's own re-ratings on accuracy (Getman et al., 2022), and automated assessment has shown promise in Korean and Mandarin-speaking children (Kim et al., 2025; Lin et al., 2024); none has yet been validated for English home practice. In a single-case study of a self-supervised home speech app, two of four children showed slight improvement on trained words, without improvement on untrained words (Strömbergsson et al., 2026). SpeechLP's acceptance model has not yet been validated against SLP ratings; that validation is the first planned study. Audio is processed on the device and never stored; only text transcripts of what the app recognized are kept.

5. Connecting Home, School, and the SLP Through Assignment and Progress Data

Parents and caregivers can play a meaningful role in speech intervention. A review of parent involvement in intervention for speech sound disorders found that parents are frequently involved in delivering home practice and that such involvement can support outcomes (Sugden et al., 2016). In related areas of communication, parent-implemented interventions have shown positive effects on children's language skills (Roberts & Kaiser, 2011). ASHA's national outcomes data found that three times as many children with severe speech sound disorders made significant gains in intelligibility when they received 21–30 sessions, twice weekly, with an SLP-directed home program (ASHA, 2024). Yet home practice is rarely measured: surveys of SLPs have not quantified how often it is assigned (Sugden et al., 2018; Tambyraja, 2020), and intervention studies seldom report what it involved (Sugden et al., 2016).

SpeechLP produces reports designed to be read by parents, teachers, and SLPs. Reports summarize screener results and the child's practice and progress over time, giving the SLP information about home practice that would otherwise be unknown, and giving families clear, concrete next steps. Because the app is guided by SLP lessons, parents and teachers do not need clinical training to support practice.

For SLPs, SpeechLP connects to **SpeechTherapist.app**, a companion web app built for the SLP workflow. SLPs invite a caregiver, assign home practice, and track games played, words practiced, and accuracy overall and by sound. Every scored practice item is logged with the word attempted, the sounds recognized, whether it was accepted, and a timestamp, so practice outside the therapy room is measured rather than assumed, and the SLP can adjust instruction based on data.

TO COMPLETE: Add a screenshot of a SpeechLP progress report or the SpeechTherapist.app progress view.



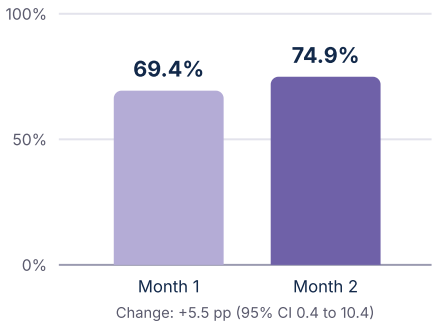
Children advance along a visual learning path and celebrate progress with the SpeechLP penguins.

Initial Evidence: A Descriptive Retrospective Study

SpeechLP published a retrospective observational study of routine practice records from January 22 through October 7, 2026 (Silver et al., 2026). The authors, all affiliated with SpeechLP, included Mari Gottlieb, M.A., CCC-SLP, and Alissa Fox, M.S., CCC-SLP, who provided clinical review. The study followed 101 returning child profiles (55 recorded as ages 2–5 and 33 as ages 6–8) across two consecutive 30-day periods and compared accuracy on identical sound targets within each profile.

What the study found

- Across 660 matched profile-and-sound pairs (11,902 practice cards), mean accepted-card accuracy rose from **69.4% to 74.9%**, a change of **+5.5 percentage points (95% CI 0.4 to 10.4)**.
- 54 of 101 profiles (53.5%) had higher second-month averages, 34 (33.7%) lower, and 13 (12.9%) unchanged.
- Of the 455 sound comparisons that changed, 61.1% increased; 205 of 660 were unchanged.
- Sixteen of 21 reportable sound means were higher in Month 2, including /k/ (+14.4 pp), /z/ (+16.0 pp), and /t/ (+11.1 pp); five were lower, including /ɹ/ (−3.3 pp).
- Practice declined in Month 2: mean sessions per profile fell from 6.73 to 3.44 and mean practice days from 4.70 to 2.71.



Mean accepted-card accuracy on matched sounds, 101 profiles (each profile weighted equally). Source: Silver et al. (2026).

How to interpret these results

These findings are preliminary and descriptive, and the study's authors state their limits directly. Accuracy reflects the app's automated speech recognition decisions, which have not been validated against SLP ratings, and scoring changed over the study period without adjustment for app version. A card counted as accepted if any attempt succeeded, so scores can reflect repeated tries and listening to the model. The cohort was selected for returning use, there was no comparison group, and many children were in an age range of active sound acquisition, so the change cannot be attributed to SpeechLP practice or interpreted as clinical articulation improvement. The study did not test untrained words or connected speech. Under ESSA, this study does not meet the requirements for Tier 3 (promising) evidence; it is reported here as part of the rationale and as the reference point for the prospective evaluation that follows.

Full manuscript and aggregate data (CC BY 4.0): speechlp.com/research/articulation-practice-two-months · https://doi.org/10.5281/zenodo.23932485

Planned Evaluation

SpeechLP is committed to building evidence of its effectiveness beyond Tier 4 and has published an invitation to university research partners. Two starting studies are proposed. In each, SpeechLP provides practice data, the app platform, engineering support, and recruitment through connected SLPs; the partner leads study design, ethics approval, and analysis. The sequence is deliberate: first validate the app's scoring against SLPs, then use the validated platform to study practice and progress.

1. Scoring validation (starter pilot)	Compare the app's scores with expert SLP ratings, acoustic measures, and parents' correct/incorrect judgments during home practice, across children with a range of speech motor skill. This replicates in English the human-versus-machine design of Getman et al. (2022), with several SLP raters instead of one and the same metrics (accuracy, unweighted average recall, mean absolute error), and adds a parent comparison. Using several raters matters because expert listening has limits of its own (Meyer & Munson, 2021; Allison et al., 2025), and combining expert listening with instrumental measures is recommended (Namasivayam et al., 2025).
2. Home practice dose (existing data)	Test whether the amount, spacing, and accuracy of home practice predict gains, and how closely families complete SLP-assigned practice, using de-identified data under a data use agreement.
Longer-term studies	Once scoring is validated: randomized comparisons of feedback type and frequency within the app; target sequencing by motor complexity compared with an expert clinician's choices; and prospective outcome studies with a comparison group and independent, functional measures such as percentage of consonants correct (Shriberg et al., 1997) and the Intelligibility in Context Scale (McLeod et al., 2012), plus stimulability and self-monitoring.
Research partner	[University partner, study selected, and timeline once confirmed]

Why generalization must be measured directly

Gains on practiced items do not guarantee carryover. In a three-child home-delivered articulation study, transfer across tested settings exceeded transfer to untrained words (Costello & Bosler, 1976). Self-monitoring has been studied as a way to support generalization, although its relationship with conversational gains is variable (Koegel et al., 1986; Shriberg & Kwiatkowski, 1990). A meta-analysis of 12 treatment studies found that treating more complex sounds also improved untreated, simpler sounds (Maggu et al., 2021), but this does not establish transfer from every form of app practice. Outcome studies will therefore measure untrained words and connected speech rather than relying on in-app scores.

Research Partnership Terms

- **Independence.** Partner researchers control the analysis and can publish regardless of results. SpeechLP reviews manuscripts within 30 days only to protect participant privacy and correct factual errors about the product, not to approve findings.
- **Agreements.** A data use agreement covers access, privacy, storage, and ownership of any models or methods a partner develops using SpeechLP data. Ethics approval runs through the partner's institutional review board.
- **Privacy.** Data is shared de-identified, with small groups suppressed. Account and child identifiers never leave SpeechLP, and audio is never stored.
- **Support.** App licenses for study participants, engineering time for data exports, and co-authorship for SpeechLP clinicians where contributions warrant it. Funding can be discussed once a specific study is scoped.
- **Transparency.** SpeechLP's cofounders have a financial interest in the app, and this is disclosed in all outputs.

About the Team

SpeechLP was founded and is led by practicing speech-language pathologists.

Alissa Fox, M.S., CCC-SLP — Cofounder and co-CEO. SLP since 2004, treating children's speech sound disorders, motor speech delays, and language disorders in schools and via teletherapy, with a focus on assistive technology. Vice President for Schools at the Iowa Speech-Language-Hearing Association; licensed in Pennsylvania and Iowa. Wrote the cueing used in SpeechLP and provides models in its video lessons.

Mari Gottlieb, M.A., CCC-SLP — Cofounder and co-CEO. Private practice owner and SLP since 2017, treating children's speech sound and language disorders in schools and via teletherapy, and cognitive-communication impairments during nearly six years at a brain injury and concussion clinic. SPEAK OUT!-trained; brings a prior advertising career to family engagement and app design; licensed in Colorado and Washington. Created SpeechLP's video lessons and provides models in them.

Shail Silver — Cofounder and Executive Chair. **Matthew Siu** and **Jabali Ndungu** — Software engineers, data and analysis support.

> **Next step.** Districts, schools, and university researchers interested in partnering on SpeechLP research are invited to a 30-minute call. Contact Alissa Fox, M.S., CCC-SLP, at **alissa@speechlp.com**.

Implementation and Instructional Support

SpeechLP runs on iPhone and iPad, so children can practice at home, in the classroom, or in the therapy room. Families can begin with the screener in minutes, and the guided learning path tells the child what to do next, minimizing setup for adults.

> SpeechTherapist.app
>
> **The SLP workflow.** SpeechTherapist.app gives speech-language pathologists one place to manage SpeechLP practice for their caseload:
>
> - **Invite and assign:** invite a caregiver and assign home practice for each child.
> - **Monitor:** track games played, words practiced, and accuracy overall and by sound.
> - **Plan:** use that data to decide what to target next and spend session time on new skills.

Privacy. SpeechLP complies with COPPA, FERPA, HIPAA, and PIPEDA. Audio is processed on the device and never stored; only text transcripts of what the app recognized are saved to SpeechLP's servers.

TO COMPLETE: Add a SpeechTherapist.app screenshot, and describe school onboarding and training, support channels, rostering/single sign-on, and device requirements.

When educators use data from digital practice to guide instruction, they can better target what each child needs next. Together, SpeechLP and SpeechTherapist.app are designed to give SLPs visibility into home practice so that session time can focus on new skills, and so that families and teachers can reinforce the same targets.

Conclusion

SpeechLP brings together several research-supported ingredients of speech sound intervention—age-based selection of target sounds, expert modeling, high-dose practice across many words, motivating game-based practice, and family involvement—in a tool children can use between SLP sessions. An initial descriptive study found higher automated accuracy scores on matched sounds in a second month of use among returning users, and the logic model and planned evaluation provide a clear rationale for its expected impact and a path to building stronger evidence of effectiveness.

SpeechLP Logic Model

PROBLEM STATEMENT

Speech sound disorders are among the most common communication difficulties of childhood, and untreated or persistent errors can affect a child's intelligibility, literacy, confidence, and participation in school. Progress depends on frequent, accurate practice, but children typically see an SLP for limited time each week, SLP caseloads are large, and many families lack access to services or do not know how to practice at home. Practice materials are often repetitive and fail to hold children's attention. SpeechLP helps close this gap with an engaging, SLP-guided practice app that identifies each child's tricky sounds and delivers personalized, high-dose practice at home or school, with progress reports that keep parents, teachers, and SLPs connected.

RESOURCES

- Age-based screener with about 15 words per age level
- Short video lessons created by Mari Gottlieb, with models by Mari Gottlieb, M.A., CCC-SLP, and Alissa Fox, M.S., CCC-SLP, using cueing written by Alissa Fox
- Library of 1,000+ target words organized by sound and position
- Word-level speech practice games (e.g., Beary Hungry, Baby Talk, Chicken Little, Pilot Penguin)
- Guided learning path with levels and rewards
- Progress reports for parents, teachers, and SLPs
- SpeechTherapist.app, the SLP companion for assigning practice and monitoring progress
- iPhone/iPad app with microphone

STRATEGIES AND ACTIVITIES

- Child completes the screener; results personalize the learning path to the child's target sounds.
- Child watches an SLP model how to produce the target sound before practicing.
- Child produces many different target words in short, game-based sessions.
- Child chooses sounds to play with and advances along a map, earning rewards.
- Parents, teachers, and SLPs review reports and reinforce the same targets.
- SLPs assign target sounds and practice in SpeechTherapist.app, then review each student's progress to plan sessions.

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OUTPUTS

- Each child's tricky sounds are identified relative to age expectations.
- Children view expert SLP models of each target sound.
- Children complete a high number of target productions per session across many different words.
- Children practice targets across many different words.
- Children complete practice sessions regularly at home and/or school.
- Parents, teachers, and SLPs receive regular progress reports.
- SLPs have current data on each student's assigned practice and progress across their caseload.

SHORT-TERM AND INTERMEDIATE OUTCOMES

Improved accuracy of target sounds in practiced words. Increased awareness of how target sounds are produced. Greater willingness to practice and increased practice frequency. Earlier identification and referral of children who may need an SLP evaluation.

As children progress through SpeechLP, they are expected to show improved accuracy of target sounds in sentences and conversation, generalization to untrained words, improved overall intelligibility, increased confidence when speaking, stronger home–school–SLP collaboration, and more efficient use of SLP session time.

LONG-TERM OUTCOMES AND IMPACTS

- Age-appropriate speech sound production and intelligibility
- Fewer residual speech errors persisting into later grades
- Shorter time to dismissal from speech services
- Reduced risk of related literacy difficulties
- Greater confidence and participation in classroom and social communication
- Increased family capacity to support communication development

ASSUMPTIONS

- An adult (parent, teacher, or SLP) is available to support the child as needed.
- Access to an iPhone or iPad with a working microphone and a reasonably quiet practice space.
- Access to the internet.
- Children with suspected speech sound disorders are also evaluated and served by a licensed SLP where available.

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Appendix A: Research Protocol

Study 1: Validating SpeechLP's Automated Speech Scoring Against SLP Ratings, Acoustic Measures, and Parent Judgments

Protocol draft, version 0.1 · November 2026 · Prepared by SpeechLP for review by prospective research partners. Items marked in yellow are proposals to be finalized with the partner.

Design	Prospective, cross-sectional measurement-validation study comparing the app's accept/reject decisions with a reference standard built from blinded SLP ratings.
Participants	40 English-speaking children aged 3–10, spanning a range of speech motor skill.
Primary outcome	Agreement between app decisions and SLP consensus: accuracy and unweighted average recall (UAR).
Lead	University partner (design, ethics approval, data collection, analysis); SpeechLP provides the app, engineering support, and recruitment through connected SLPs.
Duration	About [9] months from ethics approval to manuscript submission.

A.1 Background and rationale

SpeechLP uses automated speech recognition to decide, for each practice card, whether a child's production of the target word is accepted. In a retrospective study of 101 returning profiles, mean accepted-card accuracy rose from 69.4% to 74.9% across two months (Silver et al., 2026), but the authors could not interpret this as clinical improvement because the acceptance model had not been compared with SLP judgment. Every later study of SpeechLP's effects depends on knowing how far its scores can be trusted, so validation comes first.

A Swedish study of 28 children aged 4–10 found that a speech recognition model rated single words within about 5 percentage points of an expert's own re-ratings on accuracy, although the expert was better at distinguishing the middle levels of the scale (Getman et al., 2022). That design has not been tested in English or in home practice. Because expert listening has its own limits (Meyer & Munson, 2021; Allison et al., 2025), this study uses several SLP raters, reports their agreement with each other, and adds instrumental (acoustic) measures (Namasivayam et al., 2025).

A.2 Research questions

1. **Primary.** How closely do SpeechLP's accept/reject decisions agree with consensus correct/incorrect ratings from blinded SLPs?
2. How does that agreement compare with the agreement between individual SLPs and the consensus of the other raters?
3. Does agreement differ by target sound, word position, child age, speech motor skill, and error type (substitution versus distortion)?
4. How do parents' in-the-moment correct/incorrect judgments during practice compare with the SLP consensus and with the app?
5. For sounds with established acoustic correlates, how do app decisions relate to acoustic measures?

A.3 Participants

Inclusion. Children aged 3–10 years who speak English as a primary language, with parent or guardian consent and child assent appropriate to age. The sample will deliberately include children with identified speech sound disorders (including residual errors such as /ɹ/ and /s/ distortions) and children with typically developing speech, so that the app is tested on both correct and incorrect productions.

Exclusion. Uncorrected hearing loss, structural or neurological conditions that prevent participation in a picture-naming task, or inability to complete practice on a tablet. [Partner to confirm.]

Recruitment. Through SLPs connected to SpeechLP via SpeechTherapist.app and through the partner's clinic or school networks. Recruitment will be stratified by age band (3–5, 6–7, 8–10) and by speech status.

Sample size. 40 children, each contributing about [120] scored productions (roughly 4,800 in total). This exceeds the 28 children in Getman et al. (2022) and allows agreement to be estimated by age band and for frequently targeted sounds. A precision calculation for the primary UAR estimate, accounting for clustering of productions within children, will confirm that this sample is adequate before recruitment begins.

A.4 Procedures

1. **Baseline assessment.** A partner SLP administers a standardized single-word articulation test, from which percentage of consonants correct (PCC; Shriberg et al., 1997) is calculated, plus stimulability testing for error sounds. Diagnosis and severity are recorded where available.
2. **Practice sessions.** Each child completes two sessions of standard SpeechLP practice on a study tablet, at home or in clinic, using a fixed word list that samples all 24 consonants across word positions, weighted toward the child's error sounds.
3. **Recording.** Because SpeechLP processes audio on the device and does not store it, the partner records each session with a separate research-grade microphone under its ethics approval. The app logs each decision with a timestamp so that recordings can be aligned with app decisions at the word level.
4. **Parent judgment.** A parent sitting with the child marks each production correct or incorrect at the time of practice, without seeing the app's decision first. [Confirm whether this is captured in-app or on a form.]
5. **Blinded SLP rating.** At least three certified SLPs independently rate every production from the recordings, blinded to the app decision, the parent judgment, and each other. Each production receives a binary correct/incorrect rating and a rating on the five-level scale used by Getman et al. (2022), from unidentifiable to correct, including subphonemic errors. Raters complete a calibration set before rating, and about 20% of productions are re-presented to estimate intra-rater reliability.
6. **Acoustic analysis.** For sounds with established acoustic correlates (for example, formant measures for /ɹ/ and spectral measures for /s/ and /ʃ/), trained analysts blinded to all ratings extract acoustic measures. [Partner to specify sounds and measures.]

A.5 Measures

App decision	Accepted or not accepted for each production, with the sounds recognized and app version.
Reference standard	Majority vote of the SLP raters on the binary rating; the five-level rating is summarized as the median across raters. Uncertainty in the reference is reported, not assumed away.
Parent judgment	Correct or incorrect, recorded in the moment.
Acoustic measures	Sound-specific measures for selected targets.
Child characteristics	Age, PCC, stimulability, diagnosis and severity where available, and error type for each incorrect production as coded by raters.

A.6 Analysis plan

The analysis plan and the hypotheses below will be preregistered (for example, on the Open Science Framework) before any ratings are compared with app decisions. This addresses a limitation of the retrospective study, whose comparisons were defined after the data were inspected.

- **Primary analysis.** Accuracy and unweighted average recall (the mean of sensitivity and specificity, which is robust to the imbalance between correct and incorrect productions) of app decisions against the SLP consensus, with 95% confidence intervals from a bootstrap that resamples whole children to respect clustering. Sensitivity, specificity, and Cohen's kappa are reported alongside. Where the app produces a graded score, mean absolute error against the five-level rating is also reported, matching the metrics of Getman et al. (2022).
- **Human benchmark (RQ2).** Each SLP's agreement with the consensus of the other raters is calculated with the same metrics, giving a range of human performance against which the app is compared. Inter-rater reliability is reported as Fleiss' kappa (binary) and an intraclass correlation (five-level).
- **Moderators (RQ3).** Mixed-effects logistic regression of app–consensus agreement, with random intercepts for child and word, and fixed effects for sound, word position, age band, PCC, and error type. Sound-level results with fewer than [20] productions are reported descriptively only.
- **Parents (RQ4).** Parent judgments are compared with the SLP consensus and with the app using the primary metrics.
- **Acoustics (RQ5).** Acoustic measures are compared between accepted and rejected productions, and with the five-level SLP ratings, within each analyzed sound.
- **Decision rule.** Before data collection, the partner and SpeechLP will agree on the agreement level at which app scores are considered adequate for use as an outcome measure, for example UAR within [X] percentage points of the median individual SLP.

A.7 Ethics, privacy, and data management

- Ethics approval through the partner's institutional review board, with written parental consent and child assent.
- Study recordings are collected and held by the partner under its approval; SpeechLP does not receive audio.
- App decision logs are shared de-identified under a data use agreement. Account and child identifiers never leave SpeechLP, and small groups are suppressed in any reporting. SpeechLP complies with COPPA, FERPA, HIPAA, and PIPEDA.

A.8 Roles and independence

The partner leads study design, ethics approval, data collection, and analysis, and may publish regardless of results. SpeechLP provides app licenses, a study build that logs decisions with timestamps, engineering time for data exports, and recruitment through connected SLPs. SpeechLP reviews manuscripts within 30 days only to protect participant privacy and to correct factual errors about the product. SpeechLP's cofounders have a financial interest in the app, which will be disclosed in all outputs; SpeechLP clinicians may be co-authors where their contributions warrant it.

A.9 Proposed timeline

Months 1–2	Finalize protocol, obtain ethics approval, preregister, prepare study build and word list.
Months 3–5	Recruitment, baseline assessment, and practice sessions.
Months 5–7	Blinded SLP rating, reliability checks, and acoustic analysis.
Months 8–9	Preregistered analysis; manuscript and conference submission (for example, a co-authored ASHA Convention submission).

A.10 Study 2 (summary): Home practice dose using existing data

Questions. Do the amount, spacing, and accuracy of home practice predict gains, and how closely do families complete SLP-assigned practice? **Data.** De-identified practice records (cards, sessions, practice days, timestamps, target sounds, and SLP assignments through SpeechTherapist.app) shared under a data use agreement. **Analysis.** Preregistered mixed-effects growth models of accuracy over time, with practice exposure and spacing as predictors and starting accuracy adjusted for to limit regression to the mean; completion of assigned practice is described as the proportion of assigned sessions completed. **Interpretation.** Accuracy outcomes will be interpreted in light of Study 1's validation results. As an observational study, it identifies associations, not causes.

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